

EDUGENIE

Google Gemini Powered Learning Assistant

7. Project Documentation

Project Overview

- EduGenie is a Google Gemini-powered learning assistant with a FastAPI backend and a simple HTML/CSS frontend.
- Its documented functions include Q&A, concept explanation, quiz generation, summarization, and learning recommendations.

Prerequisites

- Python 3.10+.
- FastAPI.
- HTML and CSS knowledge for the frontend.
- A Google Gemini API key.
- Uvicorn.
- Jinja2.

Local Execution

- Install the required Python dependencies.
- Configure the Gemini API key.
- Run the application using: `uvicorn main:app --reload`.
- Open `http://127.0.0.1:8000` in a browser.

Documented Project Structure

- `main.py`, `explanation_module.py`, `qna.py`, `quiz_module.py`, `summary_module.py`, `learning_path.py`, `templates/index.html`, `static/style.css`, and `requirements.txt`.

Known Development Considerations

- The documentation identifies challenges involving AI-generated response accuracy, device performance, personalization, usability, prompt engineering, cloud integration, and data privacy.
- These considerations should be addressed through testing, prompt refinement, UI optimization, and appropriate handling of user data.

Future Enhancement Areas

- Voice-based interaction.
- Multilingual support.
- Mobile application and offline features.
- Progress-tracking dashboards.
- Gamification.
- Adaptive learning paths.
- Group study and teacher/parent dashboards.
- LMS integration such as Moodle or Google Classroom.
- Image and PDF input for doubt solving and summarization.

Source basis: EduGenie project documentation provided by the user.